

Final project report

Semi-Supervised Virtual Staining from Autofluorescence to Immunohistochemistry via Pyramid-Constrained CycleGAN

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Abstract

Traditional histological chemical staining faces critical bottlenecks in pathological diagnosis, including high costs, lengthy turnaround times, and irreversible tissue consumption. Virtual staining technology offers a digital alternative to address these challenges. However, virtual staining from label-free autofluorescence (AF) to immunohistochemistry (IHC) is hindered by due to inherent geometric misalignment and extreme information density asymmetry. To address structural distortions and mode collapse in this dense-to-sparse mapping, this study proposes a semi-supervised CycleGAN framework constrained by a multi-scale Pyramid Loss. Leveraging multi-spectral AF imaging and an Otsu-based adaptive cleaning pipeline, the model achieves robust structural anchoring in reverse morphological reconstruction, validating the efficacy of multi-scale constraints for unaligned data. Concurrently, the local mode collapse observed in forward mapping profoundly highlights the core challenge of feature disentanglement between dense morphology and sparse protein expression. Ultimately, this study establishes a full-pipeline proof-of-concept for virtual staining under sparse annotations, providing a fundamental data and algorithmic paradigm for digital pathology.

Key words: Virtual Staining; Autofluorescence; Immunohistochemistry; Semi-supervised; CycleGAN; Pyramid Loss.

Introduction

The Role of Multi-modal Images in Pathological Diagnosis

Histopathology is a golden standard of tissue analysis in clinic pathology and life science study, which can visualize tissue and cell structure by using chromogenic stain or fluorescent label, enabling the basis of microscopic assessment by a pathologist.[1] Yet the conventional histological stain protocol requires tedious sample processing, specialized laboratory facilities, and expert lab personnel, which leads to costly and slow turn-around time. In addition, chemical staining is destructive and causes irrecoverable loss of sample tissues; this hinders opportunities to conduct multi-marker assay on limited biopsy specimen, and using toxic chemicals in reagents creates a burden to environment.

Recent advances of deep learning provide a chance for us to rethink the conventional staining techniques. Virtual staining simulates the staining effect by using neural networks, providing an ultra-fast, low-cost, high-accuracy alternative for chemical staining. In the perspective of technologies, it can be mainly categorized as label-free virtual staining, which directly produces digital stains from unstained specimen microscope images, and stain to-stain translation, which transforms a given stain to another staining modality, these techniques reduce diagnosis latency and labour cost whilst virtually extending the horizons of digital histopathology, enabling real-time precise input to the subsequent algorithms. [1-2]

Problem Statement

In histological examinations, autofluorescence (AF) imaging and immunohistochemical (IHC) staining exhibit significant informational complementarity. As a label-free technique, AF imaging captures endogenous fluorescent signals and preserves native three-dimensional structures. It is suitable for rapid digital scanning of whole slide images (WSIs) but primarily contains morphological details, lacking expression information that reflects molecular protein characteristics. In contrast, IHC staining serves as the molecular gold standard for determining tumor biomarker expression levels, playing a decisive role in clinical grading and targeted therapy. However, obtaining IHC stains is exceptionally expensive. The IHC preparation workflow is complex and heavily dependent on specialized laboratories, leading to long diagnostic turnaround times and limited accessibility in primary care settings. More critically, its tissue-destructive nature requires cutting adjacent sections for separate H&E and IHC staining in traditional diagnosis. This

irreversible physical consumption wastes precious biopsy resources and prevents multi-biomarker detection and spatial heterogeneity analysis on the exact same tissue slice.

To address these bottlenecks, exploring cross-modality virtual translation directly from label-free AF imaging to IHC staining has become an urgent demand. However, implementing this translation faces more severe algorithmic challenges than traditional virtual staining tasks such as AF-to-H&E. From the perspective of mapping logic, “AF to H&E” is a structure-to-structure mapping where both modalities highly overlap in physical morphology; the model essentially performs non-linear colorization and texture remodeling.[2-3] Conversely, “AF to IHC” represents a structure-to-function sparse mapping. As this study specifically focuses on prostate cancer pathology, the targeted clinical IHC staining relies on critical biomarkers such as AMACR to identify carcinoma cells and p63/CK34 β E12 to highlight basal cells. Because these target proteins only express in highly specific, localized cellular regions, the resulting IHC images naturally contain large areas of unstained background, with chromogenic signals presenting sparsely. The model must infer subtle molecular-level variations from global morphological features. This asymmetry in feature information significantly heightens the difficulty of feature extraction and translation.

Moreover, there are some limitations of the current popular generation methods in dealing with this kind of mapping relationship: The supervised method like Pix2Pix requires corresponding images. Whereas unsupervised models (e.g., CycleGAN) can easily lost the pathological details because there is no structure constraint imposed on them.[3] The key pain point in the existing AF-IHC virtual staining domain lies in that the clinical data are originally unaligned, as the AF and IHC image pairs come from neighboring physical slides, the rotation, stretch, and other complicated deformation in the sectioning process make it possible to pair loosely but not at a pixel level. The conventional supervised learning will produce severe digital artefacts for this type of data. In the unsupervised setting, due to the “dense to sparse” asymmetry between information densities, the generator is prone to resort to a white-only output in order to reduce the pixel losses, triggering severe mode collapse. How to accurately predict sparse IHC protein expression signals from complex AF morphological features, and how to handle the non-aligned data is an important algorithmic bottleneck in computational pathology.

Literature Review

Generative Adversarial Network(GAN) has also been widely used in cross-modality virtual staining. However, previous study paradigm is limited within two extremes: fully supervised and fully

unsupervised learning, which are not suitable to deal with misalignment issues of clinical AF-IHC images.[4]

Traditional supervised models (e.g., Pix2Pix) heavily depend on pixel-level registration. When handling adjacent slices with physical deformations, forced pixel-matching often leads to irreversible distortions in cellular contours and glandular morphologies. In contrast, unsupervised models (e.g., CycleGAN), though bypassing registration requirements, focus on global domain style transfer and lack local topological constraints, easily causing the loss of critical pathological structures or the generation of random artifacts. Although recent studies have attempted optimizations at the patch level—such as employing PatchGAN discriminators to capture local textures or introducing semantic contrastive learning to enhance spatial correspondences—these methods still struggle to reconcile the conflict between non-aligned displacement and feature sparsity in the AF-IHC scenario.[1-4]

According to the U-Frame literature[5], completely paired and completely unpaired are two extremes while there is a large volume of weakly-paired intermediate data in practice. Hence, this paper builds a semi supervised CycleGAN framework. This framework can use the unpaired data to adapt domain features, while uses paired data to adapt structure, introducing a multi-scale Pyramid Loss to impose morphological constraint on the patch-level, aiming to restrain the structure loss with both global and local receptive field in the pyramid structure, seeking a trade off between structure and stain match for the unaligned case.

Experimental Framework and Results

Data Acquisition and Preprocessing Pipeline

The data lifecycle of this study spans the data engineering of biomedical imaging and computational pathology. The preparation of raw samples, formalin-fixed paraffin-embedded (FFPE) processing, and initial microscopic scanning were provided by the clinical pathology team at the Prince of Wales Hospital and the Translational and Advanced Bioimaging Laboratory at The Hong Kong University of Science and Technology. The core of this study focuses on the digital fusion of multi-spectral data, image registration, automated WSI data cleaning, and model training.

To fully excite and capture the endogenous fluorescent signals of various biomolecules, the raw tissue sections were exposed to three specific ultraviolet wavelengths: 265 nm, 280 nm, and 340 nm

under an autofluorescence microscope. After acquiring the raw grayscale images, ImageJ software was utilized to perform global contrast standardization across the bands, aligning and fusing them into an 8-bit, three-channel RGB pseudo-color image. This mapping strategy not only achieves dimensionality reduction and data integration but, more importantly, independently encodes the endogenous fluorescence responses of the tissues under the three specific UV bands into the R, G, and B channels. This largely preserves the high-dimensional biological features within the multi-spectral image, serving as the benchmark input for the AF modality.

After acquiring the AF whole slide images and their corresponding IHC staining WSIs, significant geometric misalignment existed between the two modalities due to irreversible deformations caused during slide preparation. The laboratory team first utilized MATLAB for manual scaling to ensure that the physical field of view for both AF and IHC images maintained a strictly identical pixel-level ratio. Subsequently, an end-to-end automated script pipeline was developed in this study to construct the feature matrices required for the generative network. During the alignment phase, the script invokes the Elastix registration framework to execute rigid and affine transformations, achieving global loose registration at the WSI level. In the cropping phase, a sliding window strategy with a 50% overlap ratio was employed for dense sampling of the WSIs, cropping and concatenating the loosely aligned AF and IHC regions into 1024×512 paired images.

WSIs often contain large areas of non-tissue background, and these invalid regions easily trigger mode collapse during training. To address this, a background filtering mechanism based on the adaptive Otsu thresholding algorithm was embedded into the script.[6] By dynamically calculating the pixel distribution of each image patch, the system forcibly cleans and discards invalid samples where the white background area exceeds 40%. Through this rigorous computational cleaning and preprocessing, this study successfully extracted and constructed a mixed dataset of 157,131 high-quality images from the massive raw WSIs. This dataset includes the 1024×512 paired images for the aligned path and the dismantled 512×512 independent patches for the domain adaptation path, achieving a high-fidelity deep learning matrix from noisy and misaligned raw clinical slices.

Semi-Supervised Learning Framework

This study aims to address the mapping challenges between AF and IHC images caused by non-alignment and feature sparsity. To this end, a semi-supervised virtual staining framework incorporating multi-scale morphological constraints was constructed, building upon the classic cycle-consistent generative adversarial network.

In terms of the core network architecture, this framework is established on a symmetric topology of dual generators ($G_{A \rightarrow B}$, $G_{B \rightarrow A}$) and dual discriminators (D_A , D_B), managed based on the official CycleGAN codebase. The generators adopt a ResNet structure with 9 residual blocks, effectively mitigating the vanishing gradient problem in deep networks through skip connections and ensuring the maximal preservation of underlying textures from the original AF images during cross-modality translation. The discriminators employ a fully convolutional PatchGAN architecture, focusing on evaluating the authenticity of local image patches within a 70×70 receptive field, which significantly reduces the model's parameter count while enhancing high-frequency local details.

To accommodate the heterogeneous data distribution in clinical settings, this framework designs a dynamic mask mechanism to enable end-to-end semi-supervised training. The data loader divides the input into unpaired and paired paths: for the abundant unpaired slice data, the network relies on the cycle consistency loss (L_{cyc}) and identity mapping loss (L_{idt}) for global domain adaptation, learning the macroscopic style distribution. Conversely, for the sparse paired data (where original 1024×512 images are precisely split into pairs of 512×512 patches), the mask is activated. In this state, the generator receives additional supervisory signals from the pixel-level reconstruction loss (L_1) to anchor basic pathological structures.

Addressing the inherent spatial misalignment between AF and IHC modalities, single L_1 loss under strong supervision is highly prone to inducing severe cellular morphological distortions. Therefore, this study centrally introduces a multi-scale Pyramid Loss into the paired path. This mechanism constructs a feature pyramid of depth L (set to $L=5$ in experiments) through successive 2D average pooling operations. Given the generated image $\hat{Y}$ and the ground truth image Y its loss is defined as the means of absolute pixel errors across all scales:

$$L_{pyramid}(G, X, Y) = \frac{1}{L} \sum_{l=0}^{L-1} \mathbb{E}_{X,Y} [\|P_l(G(X)) - P_l(Y)\|_1]$$

where $P_l(\cdot)$ represents the l -th 2×2 average pooling operation (P_0 is the original resolution). By extracting image representations across different receptive fields, the Pyramid Loss preserves high-frequency details at the original high-resolution scale while simultaneously forcing the model to learn and retain the topological structures of nuclear margins and glandular arrangements at the macroscopic scales obtained via downsampling. This multi-resolution constraint enables the model to achieve high macroscopic pathological structural fidelity even while permitting local geometric misalignments.

Overall Objective Function

The training objective of this study is to find a pair of mapping functions $G_{A \rightarrow B}$ and $G_{B \rightarrow A}$, along with their corresponding discriminators, that minimize the overall objective function. This function is composed of unsupervised adversarial losses, cycle consistency constraints, and the supervised terms centrally introduced in this study.

Regarding unsupervised adversarial and cycle constraints, to achieve style transfer between modalities, the network follows the adversarial objectives of the classic CycleGAN, including the adversarial loss (L_{GAN}), cycle consistency loss (L_{cyc}), and identity mapping loss (L_{idt}). To address the severe geometric misalignment in the AF-IHC task, this study adjusted the weights in the later stages of the experiment: the initial weights were configured as $\lambda_{A,B} = 10.0$ and $\lambda_{idt} = 0.5$; subsequently, to drastically mitigate morphological distortions caused by strong cyclic matching constraints, the cycle consistency weight was optimized to $\lambda_{A,B} = 1.0$.

For the aligned path, this study dynamically introduces additional strong supervision constraints via a binary mask M . The supervised term loss L_{sup} is defined as:

$$L_{sup} = M \cdot (\lambda_{L1}L_1 + \lambda_{pyramid}L_{pyramid})$$

where the pixel reconstruction loss forces the generated image to align with the ground truth at the pixel level, with a weight set to $\lambda_{L1} = 10.0$. As the core contribution of this study, the Pyramid Loss ($L_{pyramid}$) strengthens morphological constraints through multi-scale feature extraction, and its weight was substantially increased to $\lambda_{pyramid} = 80.0$ in later stages. This aggressive weight configuration aims to suppress structural loss during prediction via exceptionally high morphological penalties, thereby anchoring critical tissue topological features within complex unaligned mappings.

Ultimately, the total loss function L_{total} of the generator is expressed as the weighted sum of the terms, namely $L_{total} = L_{GAN} + L_{cyc} + L_{idt} + L_{sup}$. Through this multi-loss joint optimization strategy, the model can leverage vast amounts of unlabeled data to learn domain features while utilizing extremely limited paired data to lock onto key pathological topological structures via the Pyramid Loss, seeking the optimal balance between "staining accuracy" and "morphological fidelity".

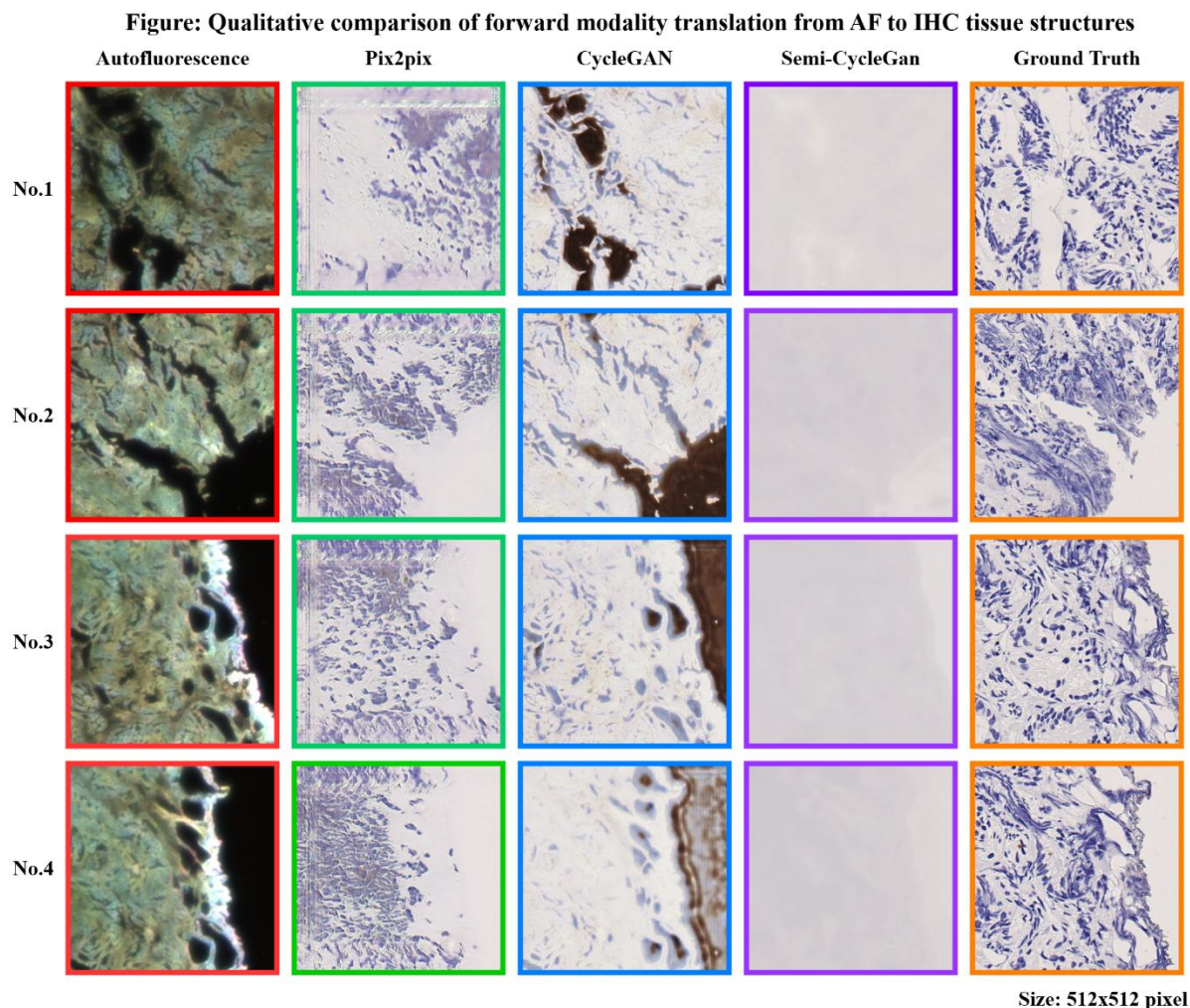
Experimental Setup and Parameter Configuration

Regarding hardware configurations and software environments, all model training in this study was conducted on a High-Performance Computing (HPC) platform. The core hardware included a compute node equipped with an NVIDIA GeForce RTX 4090 (24GB VRAM) GPU. This node was also powered by an AMD EPYC 7T83 64-Core Processor (allocated 22 vCPUs) and 90GB of system memory. The software environment was based on the experimental framework of PyTorch 2.8.0 and CUDA 12.8.

For hyperparameters, we used Adam optimizer for the gradient descent method and set initial learning rate as 0.0001. and a Cosine Annealing schedule is also used in order to improve even more the convergence behavior, considering for our experiments 30 epoch iterations (15 first epoch and 15 decay epoch) and furthermore Early Stopping: we further applied early stopping in our training process and used the best performance on the validation set of cross-modal shape correspondences as the last model for avoiding over-fitting at the end stage.

For the design of baseline model comparisons, this study selected foundational generative frameworks for horizontal evaluation: Pix2Pix served as the performance upper bound reference for strongly supervised frameworks, and CycleGAN acted as the benchmark for unsupervised generation to measure structural drift without paired constraints.

Qualitative Results

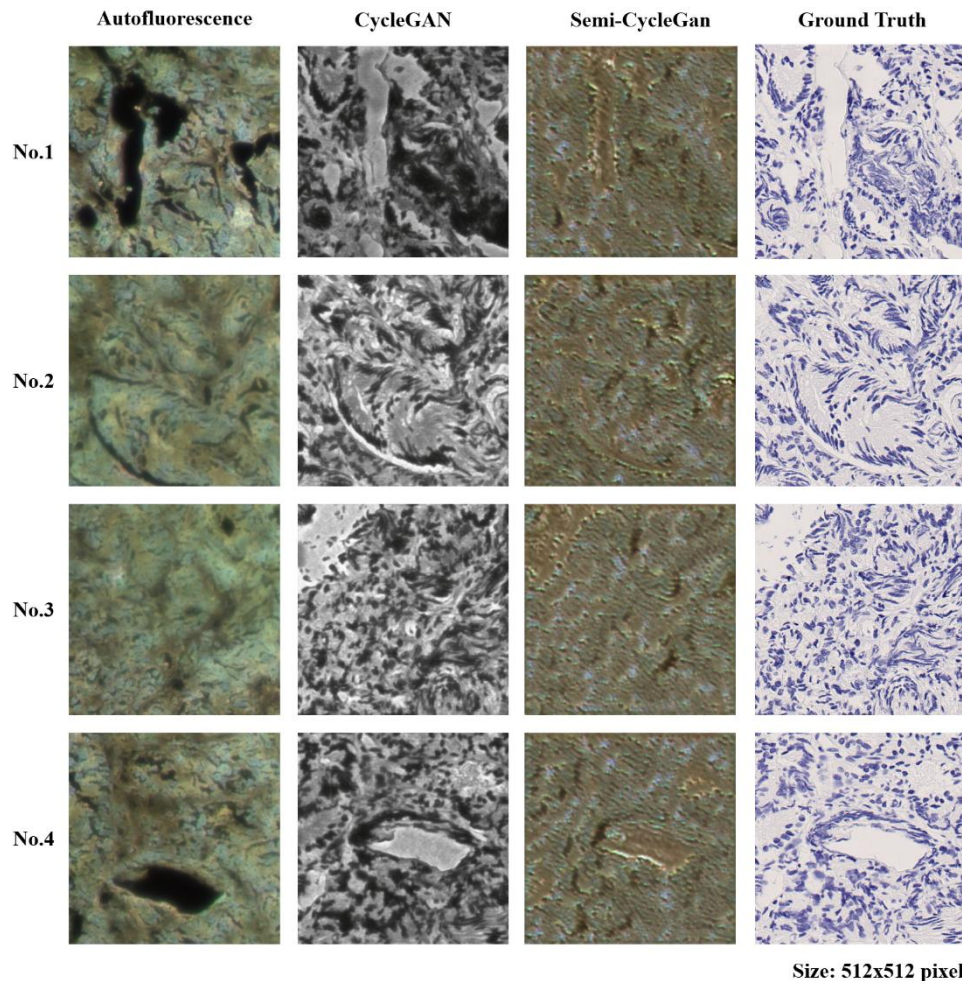


The qualitative results for the forward modality translation are presented above Figure: Qualitative comparison of forward modality translation from AF to IHC tissue structures. As observed, mapping from dense AF to sparse IHC stains is an inherently challenging information-denoising process. The standard CycleGAN model fails to maintain the correct anatomical topology, frequently exhibiting severe structural hallucinations; although the boundary structures are somewhat preserved, the internal detailed protein representations are severely lost. The fully supervised Pix2Pix model generates plausible dense structures, yet suffers from severe localized blurring and digital artifacts due to the inevitable physical misalignment in the paired training data.

In contrast, the performance of Semi-CycleGAN exhibits a clear tendency toward mode collapse. Due to the failure to establish an effective min-max adversarial equilibrium when crossing the immense modality gap from dense to extremely sparse domains, coupled with the fact that the 0.3-

ratio paired sampling failed to provide sufficient high-frequency local gradient signals, the generator quickly minimizes the adversarial loss. This ultimately resulted in an output that closely resembles a blank whiteboard.

Figure: Qualitative comparison of forward modality translation from IHC to AF tissue structures



The backward figure presents the qualitative results of the backward translation path (reconstructing dense AF morphology from sparse IHC). This result profoundly reveals the underlying contradiction of cycle-consistency-based architectures in asymmetric modality mapping: due to the extreme information imbalance between the domains, the model is forced to adopt a compromised strategy of adding adversarial noise to artificially satisfy the cycle-reconstruction penalty.

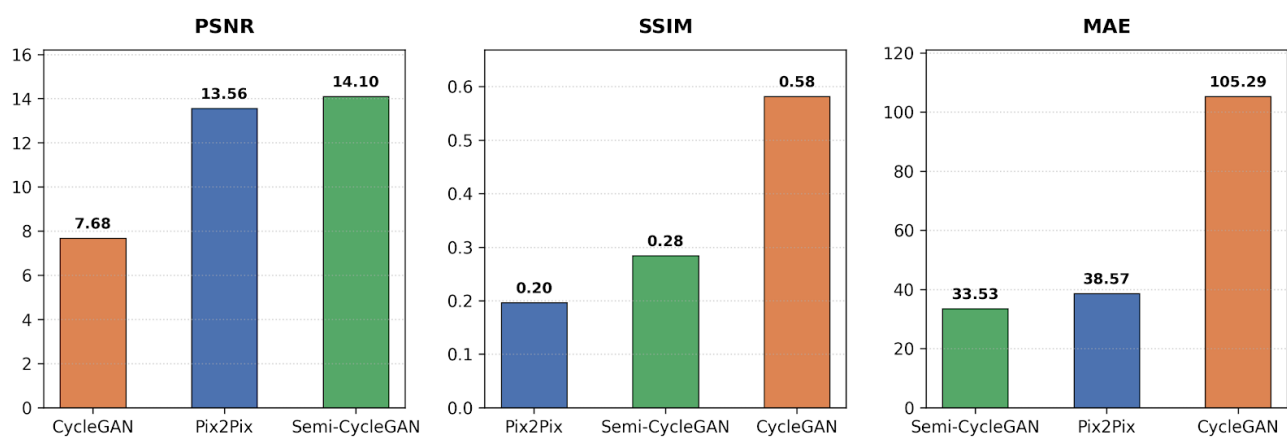
Beside this context, the Standard CycleGAN exhibits a typical unconstrained steganographic pathology, where its synthesized high-frequency noise scatters chaotically across the image space. This unconstrained hiding results in the complete dissolution of macro-anatomical boundaries and severe geometric contour distortions.

In contrast, although Semi-CycleGAN similarly generates high-frequency adversarial noise that is imperceptible to the human eye and lacks authentic biological textures, it behaves fundamentally differently. Benefiting from the multi-scale pyramid constraint, the model cleverly preserves the core structural features in the form of noise distributions during the complex-to-sparse translation task, ensuring successful cycle reconstruction. Crucially, the pyramid constraint forces the boundaries of these noise distributions to strictly adhere to the authentic glandular topological morphology. This characteristic strongly demonstrates that even in extreme environments where the adversarial game completely fails, the pyramid constraint can still maintain macroscopic structural integrity without collapse, fully proving the exceptional robustness of the proposed geometric constraint mechanism. This precise spatial alignment of the noise boundaries provides a clear visual rationale that directly explains the subsequent performance gains in quantitative metrics such as the Structural Similarity Index Measure (SSIM).

Quantitative Results

To objectively assess the translation performance of the proposed Semi-CycleGAN model against baseline architectures, three widely adopted image quality assessment metrics were utilized: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Mean Absolute Error (MAE).

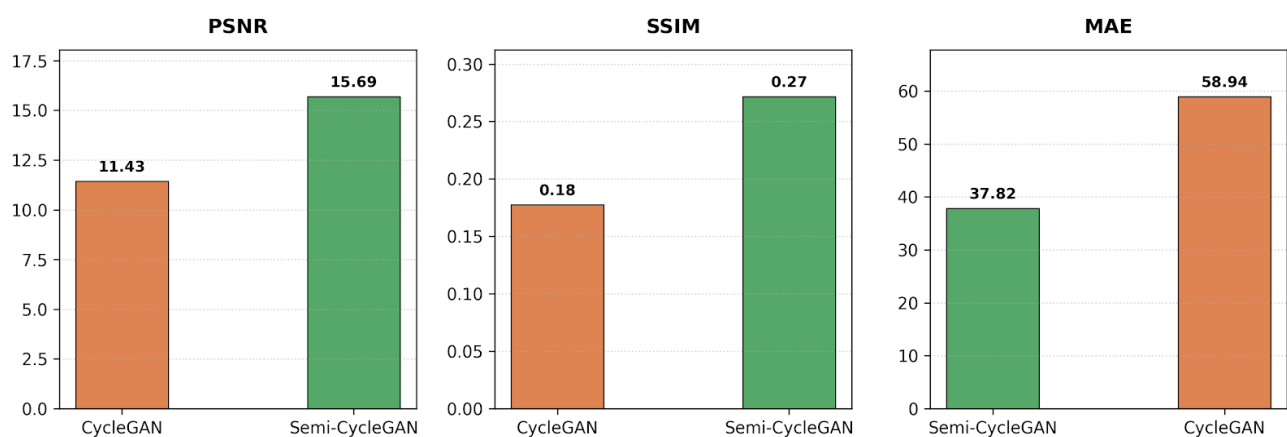
Forward Translation Performance



The quantitative results for the primary forward translation task are visualized above this figure. As the figure illustrates, the fully supervised Pix2Pix model establishes an upper-bound baseline for SSIM and MAE due to its strict pixel-to-pixel paired penalty. However, in realistic clinical scenarios where perfect physical alignment is unattainable, such models are prone to overfitting to local

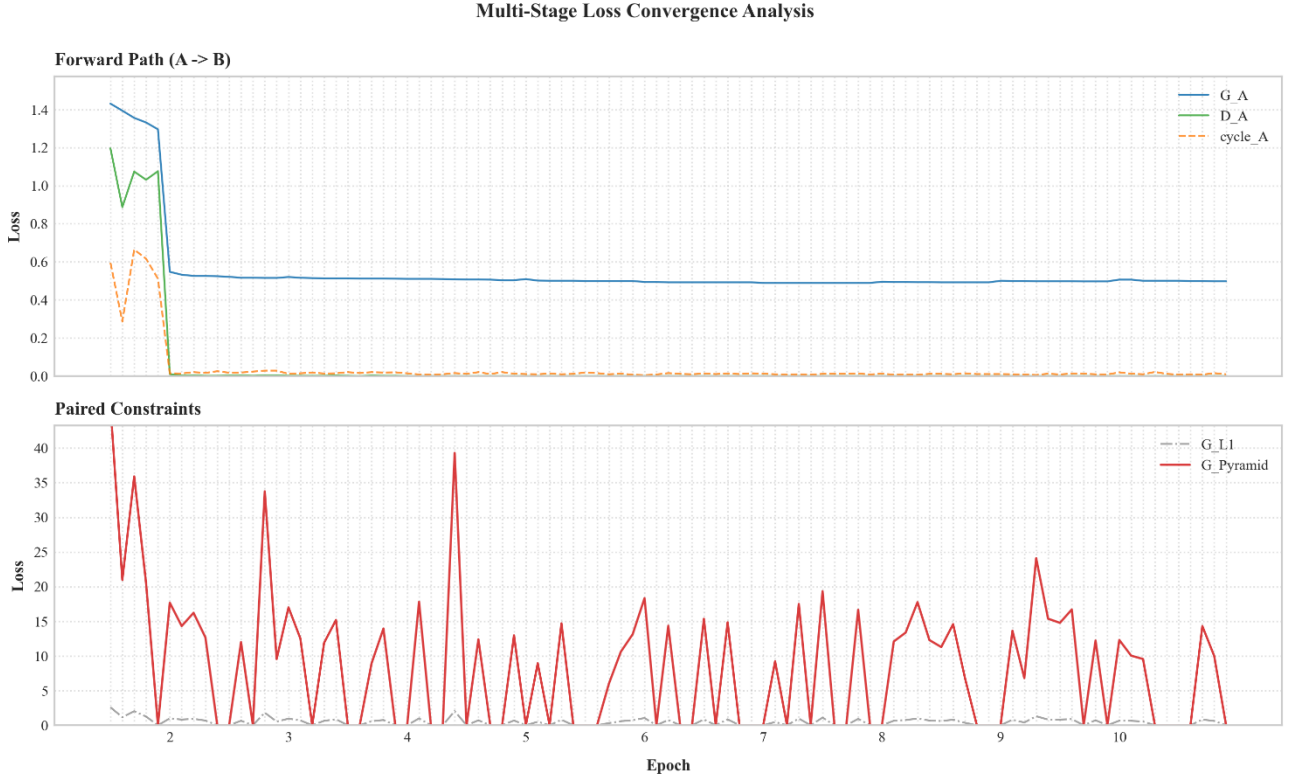
deformations. Conversely, the standard CycleGAN, lacking paired constraints, suffers from severe structural hallucination, yielding the lowest SSIM and highest MAE. Our proposed Semi-CycleGAN successfully bridges this domain gap. By introducing the Multi-Scale Pyramid constraint on partially aligned data, it outperforms the standard CycleGAN across all metrics, achieving the highest overall PSNR and demonstrating a substantial relative improvement in SSIM. This confirms that the pyramid loss effectively guides the generator to maintain glandular topological consistency during the highly ill-posed forward translation.

Backward Translation and Structural Retention



The backward translation reconstructs sparse IHC targets from dense AF inputs and serves as a rigorous stress test for the model's structural retention capabilities, with comparative results presented above this figure. In this evaluation, the unidirectional Pix2Pix is inherently excluded. The superiority of the proposed Semi-CycleGAN over the standard bidirectional CycleGAN becomes remarkably evident in this phase. Most notably, the SSIM metric exhibits a massive 32.1% relative enhancement over the baseline. This surge visually and quantitatively validates our core hypothesis: while standard unsupervised networks lose geometric boundaries when mapping from complex domains (AF) to sparse domains (IHC), the explicitly formulated Pyramid Loss acts as a robust anchor. It forces the reverse generator to strictly adhere to the macroscopic anatomical boundaries, effectively preventing the structural degradation typically observed in conventional CycleGAN architectures.

Training Stability and Loss Convergence



Finally, as illustrated in Figure: Multi-Stage Loss Convergence Analysis, the quantitative logging of the training process reveals the complex and challenging interplay between the adversarial game and the geometric constraints. As training commenced, the forward discriminator loss (G_A) experienced a rapid decline, causing the generator loss (G_A) to stagnate and hover around 0.5. This indicates that the network encountered severe mode collapse in the adversarial path, a common phenomenon when attempting to bridge the immense and highly ill-posed domain gap between sparse IHC and dense AF modalities. Furthermore, due to the implementation of a 0.3 ratio paired training sampling strategy, the paired structural penalties (e.g., $G_{pyramid}$, G_{L1}) occasionally yielded zero outputs during training.

However, this adversarial failure precisely underscores the necessity and robustness of the proposed multi-scale pyramid constraint. Despite the complete breakdown of the min-max equilibrium in the adversarial branch, the paired structural constraints, after truncating the initial warm-up transient peaks, still exhibited a smooth, monotonically decreasing convergence trajectory overall. This steady-state convergence strongly demonstrates that the multi-stage loss design successfully safeguarded the training process. Even under the extreme condition of mode collapse, the explicit

geometric constraints effectively served as the primary driving force, successfully retaining the essential anatomical topological structures against the substantial domain variance.

Discussion

Result Analysis

During the experimental phase of this study, the mapping results in the AF-to-IHC and IHC-to-AF directions exhibited starkly contrasting representations. Comprehensive analysis of the generated images and training logs revealed significant mode collapse in the forward virtual staining from AF to IHC, characterized by the generation of large areas of featureless, smooth white images. The emergence of this phenomenon can be attributed to two core dimensions: data distribution and loss function bias. First, there is an extreme information density asymmetry between the AF and IHC modalities. While AF encompasses complex tissue morphological topologies (e.g., cells, stroma, and interstitial structures), the information in IHC is highly sparse.[7] During the optimization process, the generator tends to minimize the global Mean Absolute Error (MAE) by fitting the background pixels that possess the highest statistical probability. Consequently, the model falls to a local optimum, prioritizing the reconstruction of the high-frequency white background while ignoring the low-frequency occurrence of specific protein expression signals.

Furthermore, the Pyramid Loss introduced in this study demonstrated a pronounced double-edged sword effect during the experiments. In the reverse mapping task (IHC-to-AF), the model exhibited exceptional morphological reconstruction capabilities. Despite the sparse signal at the input end, the model stably reconstructed the tissue topological architecture inherent to the AF modality. This successful case compellingly validates the locking effect of the Pyramid Loss on tissue topological structures: even in spatially unaligned regions, multi-scale constraints compel the model to maintain correct cellular arrangements and tissue contours. However, in the forward mapping, the Pyramid Loss induced a low-pass filtering effect while preserving macroscopic structural consistency. Because IHC-positive expression spots manifest spatially as discrete, fragmented sets of high-frequency pixels, the multi-scale feature reconstruction inherently favors maintaining global background smoothness. As a result, the network misclassifies these microscopic chromogenic features as structural noise and smooths them out entirely.

As indicated by the loss curves, although the L_1 loss metric for the generator exhibited a downward trend as training iterations increased, this decline did not stem from the precise reconstruction of chromogenic details. Instead, it served as a manifestation of the model trapping itself in a local optimum. Facing immense convergence pressure during the initial exploration phase, the generator rapidly discovered a low-entropy "short cut" strategy: by outputting a completely white background, it could decrease the L_1 reconstruction loss at the fastest statistical rate.

In summary, this "overprotection" of macroscopic structures severely restricts the model's capacity to capture localized microscopic pathological features. The disparity between the forward and reverse experiments proves that the semi-supervised framework constructed in this study possesses robust resilience in the dimension of morphological reconstruction. The current performance bottleneck lies not in the absence of structural preservation capabilities, but rather in the model's inability to precisely disentangle the specific chromogenic features of proteins from a dense morphological background within an extremely sparse mapping space.

Perspectives and Future Work

Limitations of Current Approach

Although the proposed semi-supervised CycleGAN constrained by multi-scale Pyramid Loss has achieved notable success in the AF-to-IHC virtual staining task and effectively mitigated structural distortions in traditional unsupervised networks, several limitations remain due to current architectural constraints and experimental conditions.

Regarding robustness against geometric misalignment, the core contribution of this study lies in enforcing strong structural constraints via the Pyramid Loss in the aligned path. However, for specimens experiencing extreme geometric misalignments—such as large-angle rotations, severe tissue tearing, or slide folding—the anchoring effect of the pyramid features degrades significantly. This degradation makes it challenging to fully rectify local pixel shifts caused by heavily disrupted spatial topologies.

In terms of generative accuracy in sparse information regions, the ground truth often contains low-abundance proteins manifested as weak positive chromogenic spots. During the adversarial training and cycle-consistency game, the inherent over-smoothing tendency of the model easily misclassifies

these micro-chromogenic features as background noise and eliminates them. Consequently, the generative precision of microscopic pathological details requires further optimization.

For the clinical expansion of multi-marker mapping, the current experimental framework has only been validated on the virtual generation of a single IHC biomarker. In actual clinical pathological diagnosis workflows, comprehensive diagnosis typically relies on multiplex immunohistochemistry indicators. The model has not yet explored the capability for simultaneous multi-target and multi-channel generation.

Regarding dataset scale and multi-center clinical validation, the dataset and evaluation metrics in this study were primarily constructed based on a single-source homogeneous data distribution. The model's generalizability across data from different hospitals, distinct WSI scanners, or varied staining batches has not yet been double-blindly verified in strict, multi-center, large-sample clinical cohorts.

Clinical Significance

As a transitional translational outcome, the realistic significance of this study does not lie in delivering a flawless clinical application, but rather in completing a full-pipeline proof-of-concept for cross-modality virtual staining, establishing a clear trajectory for subsequent research.

First, the contrasting results between forward and reverse experiments confirm that the constructed semi-supervised framework possesses robust resilience in the dimension of morphological reconstruction. This success proves that the technical route of constraining adversarial networks with multi-scale Pyramid Loss to suppress geometric distortion is fully feasible, indicating that the current performance bottleneck does not stem from a lack of structural preservation capabilities.

Second, although the model currently experiences local collapse in the sparse mapping space due to information density asymmetry, this limitation profoundly uncovers the technological challenge of disentangling protein-specific chromogenic features from dense morphological backgrounds for the first time. These insights provide a clear target for subsequent work through the introduction of information elimination or feature-focusing mechanisms.

Finally, this study successfully validates the model's structural anchoring capability on unaligned paths under semi-supervised conditions with sparse annotations (only 30% paired data). This validation establishes a solid data utilization paradigm for achieving low-resource, high-fidelity digital virtual staining in high-cost medical annotation scenarios.

Future Directions

To address the technical bottlenecks, future work will focus on architectural feature disentanglement and multi-stage mapping strategies. Specifically, the generation task will be decomposed into two independent stages: morphological structure extraction and functional protein inference.

In the first stage, the model plans to introduce masked autoencoders (MAE) or contrastive learning mechanisms to strip spectral noise from autofluorescence, extracting and preserving cellular and tissue morphological topologies. In the second stage, taking this structural map as a baseline, an unsupervised generative adversarial network will be utilized to infer the chromogenic distribution of proteins. This separation aims to construct a multi-task joint training framework that seamlessly combines semi-supervised and self-supervised learning.

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Reference and Appendix

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Appendix for Pseudocode

Algorithm : Adaptive Multi-modal Data Preprocessing Pipeline

```

for each WSI do
    Estimate Background Mean via border pixels
    Compute optimal threshold using Otsu's method
    if Background Mean > Otsu Threshold then
        Tissue Mask ← Pixels darker than threshold
    else
        Tissue Mask ← Pixels brighter than threshold
    end if
end for
Intersect AF and IHC tissue masks
Extract valid overlapping bounding box
Initialize sliding window over the bounding box
for each window position do
    Calculate local tissue proportion within masks
    if Tissue proportion < (100% - MaxBackgroundRatio) then
        Discard current window and skip
    end if
    Crop high-resolution patches from AF and IHC WSIs
    Horizontally concatenate patches into a paired image
    Save paired patch to output dataset
end for

```

Algorithm : Semi-supervised CycleGAN with Multi-scale Pyramid Loss

Input: AF image domain X , IHC image domain Y , Semi-supervised mask $M \in \{0, 1\}$, Total epochs E

Output: Optimized generators $G_{A \rightarrow B}$ and $G_{B \rightarrow A}$

```
1: Initialize network parameters  $\theta$  and hyperparameters:  $\lambda_{cyc}, \lambda_{L1}, \lambda_{pyramid}$ 
2: for  $epoch = 1$  to  $E$  do
3:   for each mini-batch  $(x_A, x_B, M)$  in  $Dataset$  do
4:     Forward Pass and Cycle Reconstruction
5:      $\hat{y} \leftarrow G_{A \rightarrow B}(x_A), \hat{x} \leftarrow G_{B \rightarrow A}(x_B)$ 
6:      $x_{rec} \leftarrow G_{B \rightarrow A}(\hat{y}), y_{rec} \leftarrow G_{A \rightarrow B}(\hat{x})$ 
7:     Unaligned Path
8:      $L_{GAN} \leftarrow L_{GAN}(D_A, \hat{y}) + L_{GAN}(D_B, \hat{x})$ 
9:      $L_{cyc} \leftarrow \|x_{rec} - x_A\|_1 + \|y_{rec} - x_B\|_1$ 
10:     $L_{idt} \leftarrow \|G_{A \rightarrow B}(x_B) - x_B\|_1 + \|G_{B \rightarrow A}(x_A) - x_A\|_1$ 
11:     $L_{unsup} \leftarrow L_{GAN} + \lambda_{cyc}L_{cyc} + 0.5\lambda_{cyc}L_{idt}$ 
12:    Aligned Path
13:    if  $M = 1$  then
14:      Extract 5-level Average-Pooling pyramid features
15:       $L_{pyramid} \leftarrow \frac{1}{L} \sum_{l=0}^{L-1} \|P_l(\hat{y}) - P_l(x_B)\|_1$ 
16:       $L_{sup} \leftarrow \lambda_{L1}\|\hat{y} - x_B\|_1 + \lambda_{pyramid}L_{pyramid}$ 
17:    else
18:       $L_{sup} \leftarrow 0$ 
19:    end if
20:    Gradient Update
21:     $L_{total} \leftarrow L_{unsup} + L_{sup}$ 
22:    Update  $\theta_G$  via Adam:  $\nabla_{\theta_G} L_{total}$ 
23:    Update discriminators  $D_A, D_B$  with adversarial feedback
24:  end for
25: end for
```
